

## ART AND ILLUSION

Thursday, 9 April 2026

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**ART IS ILLUSION.** A painting or drawing is a flat surface covered with marks, yet those marks can be seen as bodies, faces, reflections, distance, glass, metal, or skin. The painted or drawn mark is never the thing itself. Art works because it prompts the viewer to experience marks as if they were real objects.

A painter of realism therefore does not simply copy what is seen. They construct an image that guide perception. Linear perspective, tonal modeling, reflected light, shadow, and optical detail are not ends in themselves. They are methods for causing the mind to assemble a convincing visual experience from marks on a plane.

Any painting or drawing can only partly succeed as an illusion. A flat image cannot reproduce the full richness of embodied sight. It cannot literally give us solid bodies. What it can do is evoke, in the mind of the viewer, an experience analogous to seeing real things.

This connection between art and illusion was recognized long ago. Pliny understood that seeing is not accomplished by the eye alone. The eye receives light, but the mind is the true instrument of vision. What we see is not merely what falls upon the retina. It is what the brain makes from the incoming pattern of light.

the mind is the real instrument of sight and observation, the eyes act as a sort of vessel receiving and transmitting the visible portion of the consciousness.

**EVERY DRAWING IS ALREADY AN ILLUSION.** Real objects are not bordered by actual black lines. What we call an outline is a construction of the visual system. The eye registers differences in light reflected from adjacent regions of the world, and the brain converts those differences into perceived boundaries. A line placed at the edge of a form does not imitate its physical structure, but the way objects are segmented by vision.

Thus, drawing begins naturally with outline. To draw the contour of a face, a hand, or an eye leverages a basic operation of visual perception. The line is persuasive because the brain is disposed to treat boundaries as evidence of objects. The human brain transforms patterns of illumination – what happens when light is projected into the eye and onto the retina – into what we see.

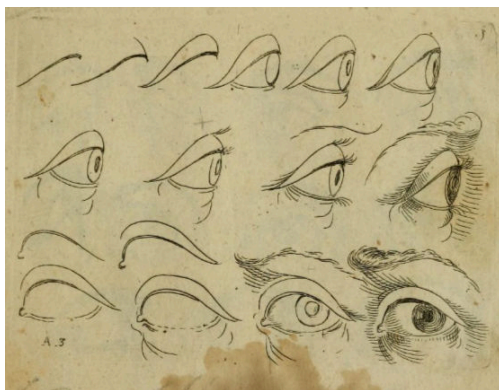


Figure 191: Odoardo Fialetti's drawing manual reduces the eye to a sequence of contours. The image works because the visual system readily accepts boundary lines as sufficient evidence for form.

WE SEE OBJECTS ABOVE ALL BY SEEING EDGES. A boundary in the world often appears where there is a sudden change in light intensity between one region and the next. Within a single object, light may vary gradually across a surface. At the object's perimeter, by contrast, the change is often abrupt. The visual system treats such discontinuities as signs of separate things.

This is why black-and-white photography works so well. A monochrome photograph removes color altogether, yet the world remains legible. Trees, sky, water, stone, shadow, and reflection are still recognizable because objects are not primarily recognized by color. Objects are defined by contrast, contour, and the organization of regions by light.

A photograph in black and white demonstrates a fundamental principle. Vision does not require a complete copy of reality in order to recognize reality. A sufficiently organized visual pattern is enough.



Figure 192: Carleton Watkins, Yosemite. Even without color, the image remains fully intelligible because the visual system identifies objects through contrast, edge, and spatial organization.

THE PHYSIOLOGY OF VISION helps explain why line and contrast are so powerful. The retina is not a neutral recording surface like photographic film. It is a sheet of neural tissue that begins the work of interpretation before visual signals ever reach the brain.

Santiago Ramón y Cajal, the great anatomist of the nervous system, was the first to map the cellular organization of the retina in detail. His drawings show extensive lateral connections among neighboring regions. We now understand that such circuitry contributes to edge detection. When one area of the retina is strongly stimulated by light, nearby regions are suppressed. The net effect enhances local contrast at boundaries.

This process is often described as center-surround inhibition. It allows the visual system to register not how much light is present, but where light changes sharply across space. The retina emphasizes places where one object ends and another begins.

This is why a drawn line can be compelling. From the beginning, the visual system is converting contrast into contour. Once an boundary is established, the mind supplies its form.

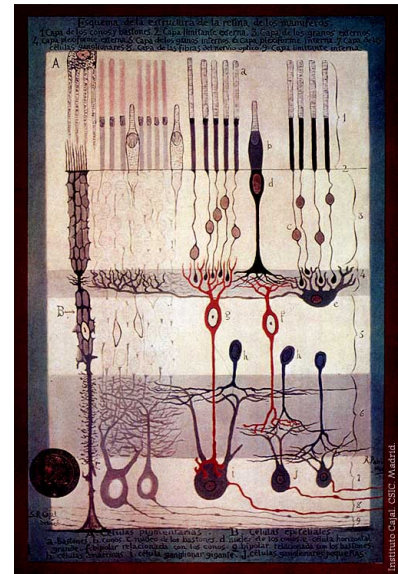


Figure 193: *Structure of the Mammalian Retina* c.1900 By Santiago Ramon y Cajal. a. Rods. b. Cones. c. Rod nucleus. d. Cone Nucleus. e. Large horizontal cell f. Cone-associated bipolar cell. g. Rod-associated bipolar cell. h. Amacrine cells. i. Giant ganglion cell. j. Small ganglion cells.



Figure 194: An edge filter in computer vision that is based on lateral inhibition like the retina. [From Wikipedia.](#)



THE VISUAL SYSTEM DOES NOT MERELY DETECT BOUNDARIES. It also exaggerates them. One consequence is the phenomenon known as Mach bands. When adjacent regions of uniform brightness meet at a sharp boundary, the darker side appears slightly darker near the edge, while the lighter side appears slightly lighter. These bands are not present in the stimulus itself. They are generated by the operations of the visual system.

Mach bands matter because they show that seeing is not passive but involves active computation. The visual system heightens contrast at boundaries in order to clarify structure, and in doing so it also produces visual illusions.



Figure 195: Mach bands. The apparent darkening and lightening near each boundary is generated by visual processing, not by actual bands painted into the image.

Artists can exploit this fact. Jesse Stillwell, for example, uses sharp juxtapositions of flat color in ways that cause the eye to perceive gradients that are not literally painted. The painting works because the visual system cannot help but follow its own neural rules.

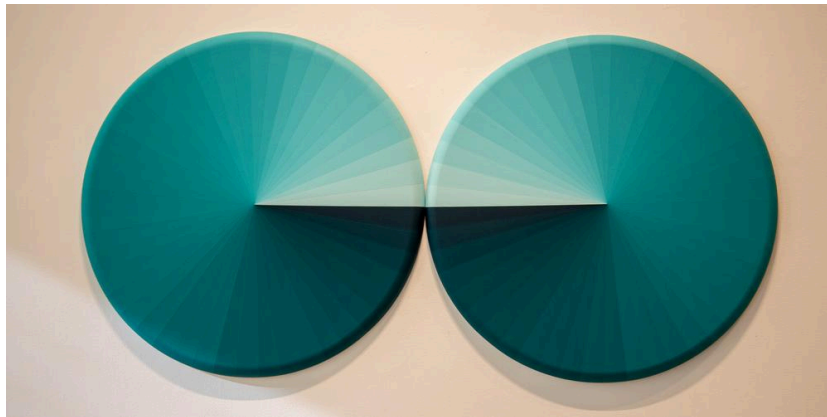


Figure 196: *Smack Dab in the Middle of Spectrum* by Jesse Stillwell, 2022. Acrylic on shaped canvas. 30"x60"

What these examples show is that visual experience is constructed at many levels. The mind draws boundaries where there are changes in light. It exaggerates contrast at those boundaries. It organizes incomplete information into coherent objects. A painting succeeds not when it copies everything, but when it gives the visual system enough information to complete the world for itself.

Johannes Vermeer's *Girl with a Pearl Earring* offers another kind of illusion. The painting is often described as a *tronie*, not a conventional portrait of a known individual but a study of expression, costume, and type. Whether the girl corresponds to any real sitter matters less than the fact that Vermeer makes us experience a convincing human presence through remarkably selective means.

The image appears full and immediate, yet Vermeer does not describe everything equally. He withholds as much as he gives. He paints only those cues required to give the impression of a face turning in light, of moist eyes, lips, skin, and a luminous pearl suspended at the ear.

Realism in painting does not depend on exhaustive transcription. It depends on the precise placement of cues to which the visual system is especially responsive.



Figure 197: Johannes Vermeer, *Girl with a Pearl Earring*, c. 1665, Mauritshuis, The Hague.

High-resolution imaging makes Vermeer's economy of means clear. The pearl is not rendered by minute descriptive detail. Vermeer suggests it with only a few strategically placed strokes: a bright highlight, a softer reflection below, and the surrounding tonal context that allows the eye to read the pearl as a smooth, rounded, and lustrous object.

The mind completes the pearl. It is seen not because every optical fact has been copied, but because enough have been supplied that the mind can complete recognition.



Figure 198: Detail of the pearl in Vermeer's *Girl with a Pearl Earring*. The pearl is not described point by point; a few strategically placed highlights and tonal relations are enough for the mind to recognize a smooth, lustrous sphere.

The same is true of the nose. Look carefully at the bridge of the nose and the transition into the cheek. Vermeer does not paint a hard contour. The form is inferred from adjacent shadow and light. The viewer sees a nose where painted information is surprisingly absent.

Technical analysis confirms how deliberately these effects were built. Imaging reveals underlayers, fine contours, and revisions that helped Vermeer control the final appearance of surfaces and edges. But the crucial lesson is simple: the painting works because the viewer does not passively receive images of objects. The viewer completes perception by active mechanisms.

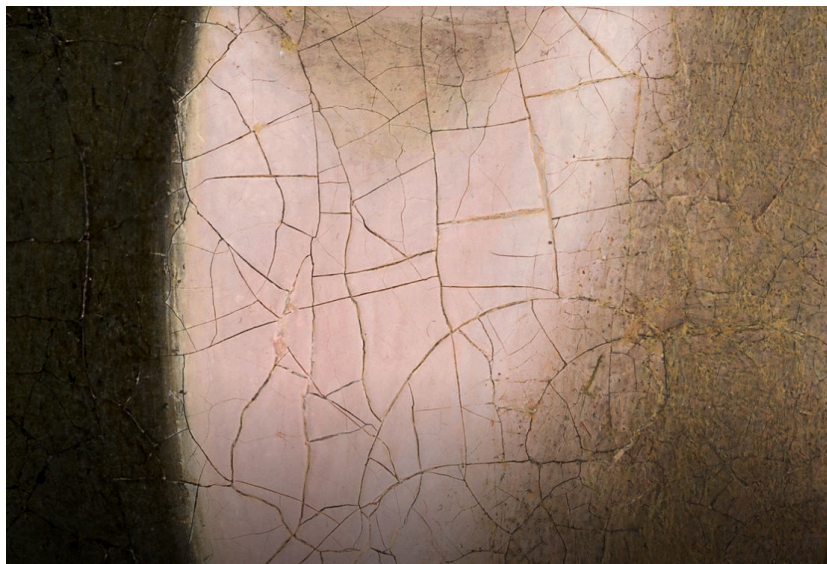


Figure 199: Detail of the bridge of the nose. Vermeer avoids a hard contour, allowing adjacent light and shadow to imply projecting form. The nose is perceived through inference as much as through explicit drawing.

Digital imaging and analysis reveals how Vermeer made his painting: layer by layer, correction by correction, with extraordinary patience. He did not simply place a final image onto the canvas in one act of transcription. He built the illusion slowly. Vermeer laid down multiple underlayers that varied in tone, allowed them to dry, and then returned to them with thinner or thicker upper layers in different regions of the painting. The final effects of light and shadow were accumulated through a sustained process.

High-resolution imaging also reveals preparatory and corrective work beneath the finished surface. Fine black outlines helped establish contours in advance, and pentimenti show that Vermeer adjusted the image as he worked, relocating features such as the iris and the ear. These are not incidental corrections. They show a painter repeatedly testing what the eye would see, then refining the image until the desired perceptual effect was achieved.

Geometric analysis makes the same point at another level. The reflections in the eyes and pearl, and the shadow cast by the nose, all align with the same incoming direction of light. Such consistency is not accidental. It shows Vermeer checking the internal logic of illumination across the whole picture. The painting's immediacy is itself deceptive, containing hidden evidence of obsessive construction.

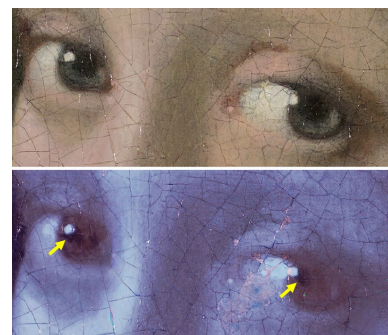
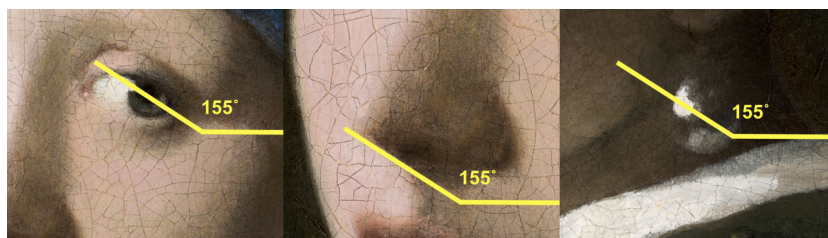


Figure 200: Evidence for pentimenti in the Girl's eyes. a Visible light photograph. b MS-IRR false colour detail. Dark marks indicate possible earlier iris locations (yellow arrows)

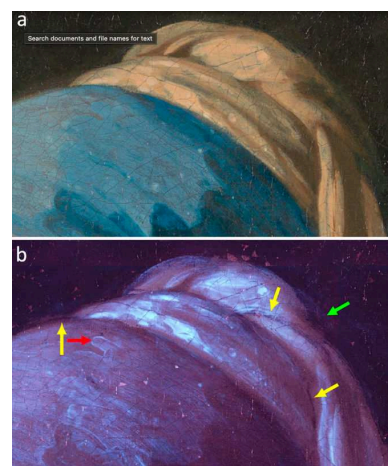


Figure 201: Evidence for contours and fine black outlines in the top of headscarf. a Visible light photograph. b MS-IRR false colour detail. Wavy brushstrokes at the surface (red arrow), fine black outlines applied in a preparatory phase (yellow arrows), back of the headscarf applied on top of an infrared-absorbing layer (green arrow).



Before reading further, look carefully at the image below. In this orientation it appears to be a conventional still life: an arrangement of edible things laid out for inspection.

Do not turn the book yet. Let the image first remain what it appears to be.

Then turn the page upside down.





The still life has become a face! Nothing has been added to the picture. No new contour has appeared, no new color has been painted, no new object has been added. Only the orientation has changed, and with it the brain's interpretation of the same forms.

Arcimboldo makes visible a deep fact about vision. The eye does not merely deliver a list of local features. The visual system searches constantly for coherent wholes, and among the most privileged of those wholes is the human face.



Figure 202: Giuseppe Arcimboldo, *The Vegetable Gardener (L'Ortolano)*, ca. 1587–1590, Museo Civico Ala Ponzone, Cremona. In this orientation the same painted forms are reorganized as a face. Without adding a single new mark, the still life becomes a portrait, demonstrating that perception does not merely register parts but actively organizes them into a coherent whole.

Arcimboldo's paintings allow two readings at once. We can inspect them part by part, recovering individual objects. But we can also see them holistically, as a face that appears all at once. The whole becomes more than the sum of its parts.

Pictorial illusion is not a matter of imitating light, texture, or perspective. It is a matter of activating the mind's cognitive processes. A demonstration of the principle that Arcimboldo leverages is the so-called Thatcher illusion. In this illusion, a face is manipulated by turning the eyes and mouth upside down within the head. When the entire image is itself inverted, the distortions are strangely difficult to detect. But when the head is returned to its normal upright position, the face suddenly appears grotesque.



Figure 203: The Thatcher illusion. When the whole face is inverted, grotesque inversions of the eyes and mouth are surprisingly difficult to detect. When the head is returned to its normal upright orientation, the distortions become immediately apparent. Upright face perception is therefore unusually sensitive to the configuration of features.

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